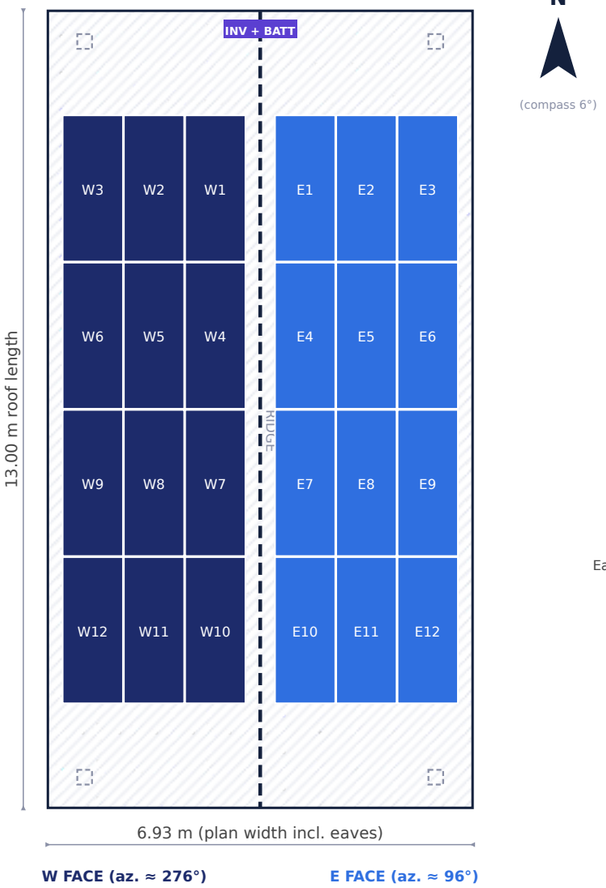
SOLAR ESS
Solar Energy for Sustainable Solution

Solar ESS Co. Ltd. | 53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694, 089 632 5829 | www.solaress.org | www.โซลาร์เอสส์.com

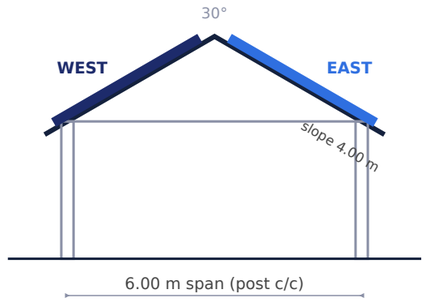
PAVILION ROOFTOP – MAE SUAI DAM AREA: PV ROOF LAYOUT, 24 × LONGi Hi-MO X10 ANTI-GLARE

Open pavilion, gable roof, ridge N–S · 12 modules on East face + 12 on West face · landscape, 3 rows × 4 columns per face · 24 × 650 W = 15.60 kWp

① ROOF PLAN (TOP VIEW) – SCALE 1:100

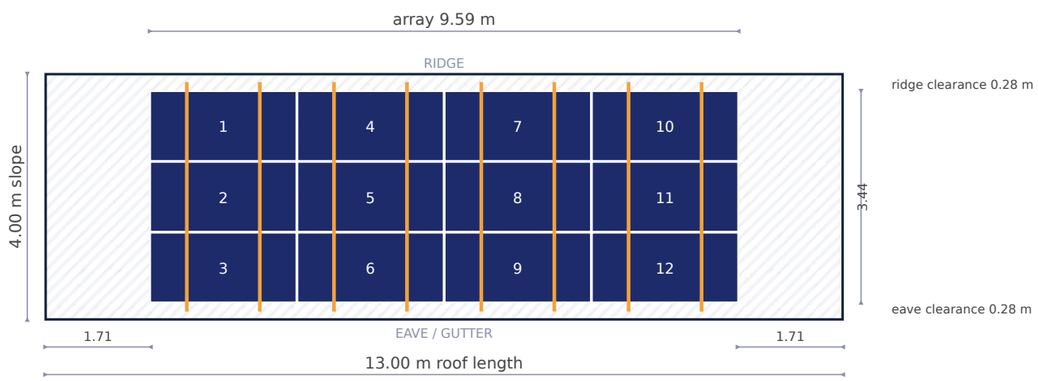


② GABLE END SECTION (LOOKING NORTH) – 1:125



Eave overhang ≈ 0.46 m each side · module plane ≈ 0.12 m above tile

③ EAST / WEST FACE – TRUE SLOPE VIEW (both faces identical, mirrored) – 1:100



MODULE & ARRAY DATA

Module	LONGi Hi-MO X10 Anti-Glare 650 W
Model	LR7-72HVH-650M
Size (L×W)	2382 × 1134 mm
Quantity	24 (12 E + 12 W)
Array power	15.60 kWp
Orientation	Landscape, flush to roof
Roof slope	4.00 m per side (pitch 30° assumed)
Per face	3 rows × 4 columns
Array per face	9.59 × 3.44 m
Rails	16 × ~3.6 m (2 per column)

STRINGING (one MPPT per face)

String A – East face	E1–E12 (12 in series) → MPPT 1
String B – West face	W1–W12 (12 in series) → MPPT 2
Inverter	Solis 16 kW 1-phase hybrid
Battery	4 × Solis IntelliHome 16 kWh
Equipment	North gable, centre (enclosed bay)

NOTES

Roof slope 4.00 m per side (site). Length 13.0 m and pitch 30° estimated from photos (5 posts/side) – verify before ordering.
East and west faces must not share one MPPT.
Check 12-module string Voc at min. site temp. against inverter max DC input before final stringing.
Hooks/rails fix to steel purlins, not the roof sheet; set rail positions before ridge cap & flashing go on.
Array centred on slope: ≈0.28 m clear at ridge and at eave.
Clamps on long module edges, within LONGi clamp zone.

Sketch for layout planning only – dimensions to be confirmed by site measurement.

PROJECT	SITE	DRAWING	DWG NO. / REV	DATE	BY
Pavilion rooftop PV + battery (job post)	**Mae Suai Dam area, Mae Suai, Chiang Rai**	**PV Roof Layout – Sketch**	**JOB-PV-01 / Rev02**	**08 Nov 2026**	**Solar ESS Working Group**